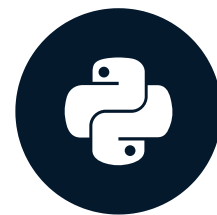


Welcome to Python!

PYTHON FOR MATLAB USERS



Justin Kiggins
Product Manager

Prerequisite: Manipulating matrices in MATLAB

```
% Example of manipulating matrices in MATLAB
```

```
v = [16 5 9 4 2 11 7 14];  
disp(v(5:end))
```

```
A = magic(4);  
disp(A(2:4,1:2))
```

```
A(A>12) = 10;
```

```
disp(A(:,1))
```

Prerequisite: Plotting data in MATLAB

```
% Example of plotting data in MATLAB
```

```
figure
```

```
plot(t,y,'b-')
```

```
xlabel('Time (s)')
```

```
ylabel('Sensor A')
```

```
figure
```

```
scatter(y1,y2,'go')
```

```
xlabel('Sensor A')
```

```
ylabel('Sensor B')
```

```
figure
```

```
histogram(y1,[0:0.01:1])
```

Prerequisite: Control flow of MATLAB scripts

```
% Example of control flow in MATLAB

fid = fopen('magic.m','r');
count = 0;
while ~feof(fid)
    line = fgetl(fid);
    if isempty(line) || strncmp(line,'% ',1) || ~ischar(line)
        continue
    end
    count = count + 1;
end
count
```

If you don't know MATLAB...

Introduction to Python

Introduction to Data Science in Python

Python for R Users

Introduction to Python for Finance

Python does more than Data Science

- General purpose programming language
 - Integrate machine learning models into large-scale applications
 - Query data from public APIs
 - Build websites that can handle large traffic volumes

Getting started with data types

- Integer
- Float
- Boolean
- String

```
# Integer  
x = 1  
print(x)  
type(x)
```

```
1  
<class 'int'>
```

```
# Float  
x = 1.0  
print(x)  
type(x)
```

```
1.0  
<class 'float'>
```


Mathematical operators

Operation	Python Operator
Addition	+
Subtraction	-
Multiplication	*
Division	/
Exponentiation	**

```
a = 3 + 12  
print(a)
```

```
15
```

```
b = 4 * 5.0  
print(b)
```

```
20.0
```


$$area = \pi r^2$$

```
radius = 5
pi = 3.14
area = pi * (radius ** 2)
print(area)
```

```
78.5
```

Warning: Do NOT use the caret operator, `^`

```
# This won't take 4 to the second power
print(4 ^ 2)
```

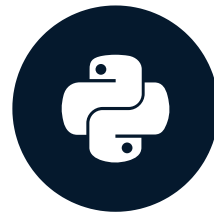
```
# This is the bitwise XOR
6
```

Let's get started!

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Methods and packages

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Methods are functions attached to variables

- Specific to the variable type
- Accessed with a period

```
yelling = "WHY ARE YOU YELLING?"  
whispering = yelling.lower()  
print(whispering)
```

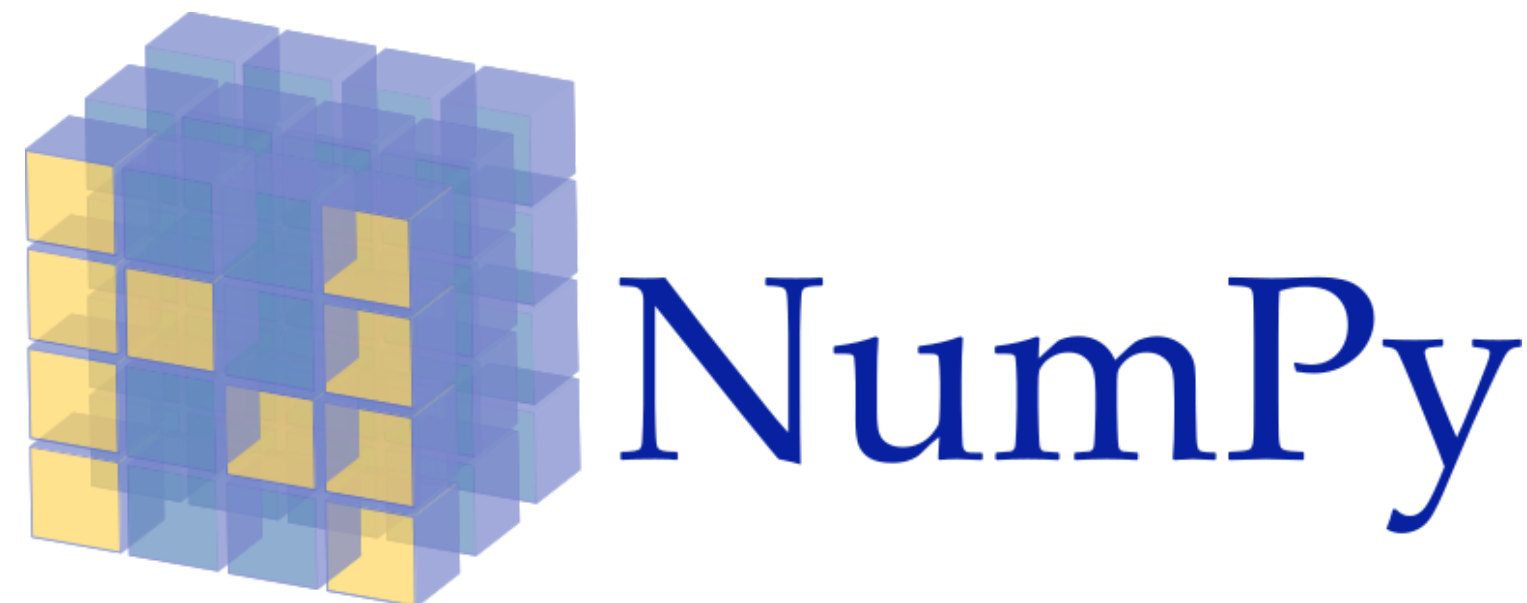
```
why are you yelling?
```

The .format() method

```
n_avocados = 1765  
message = "I have {} avocados!".format(n_avocados)  
  
print(message)
```

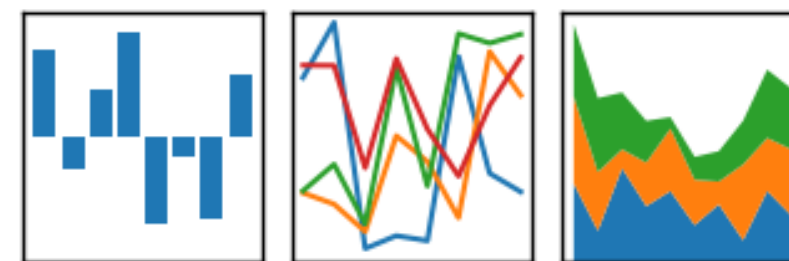
```
I have 1765 avocados!
```

Python packages



pandas

$$y_{it} = \beta' x_{it} + \mu_i + \epsilon_{it}$$



```
import math  
math.pi
```

```
3.141592653589793
```

```
math.log(0.9)
```

```
-0.10536051565782628
```

Importing part of a package:

```
from math import log  
log(0.9)
```

```
-0.10536051565782628
```


Package aliasing

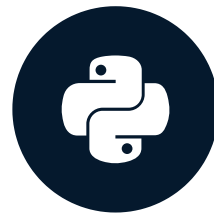
```
import datetime as dt
birth_date = dt.datetime(1961,8,4)
# Import NumPy package with common alias
import numpy as np
x = np.ndarray([1, 2, 3])
# Import pandas package with common alias
import pandas as pd
df = pd.DataFrame()
# Import Matplotlib pyplot module with common alias
import matplotlib.pyplot as plt
plt.show()
# Import Seaborn package with common alias
import seaborn as sns
sns.set()
```


Let's explore some useful methods and packages

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Arrays & plotting

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NumPy arrays

- Arrays = matrices
- Arrays can be multidimensional
- Every element is the same type

Indexing into NumPy arrays

```
print(arr)
```

```
[[2, 4, 8, 16, 32]]
```

- Zero-based indexing

```
print(arr[2])
```

```
8
```

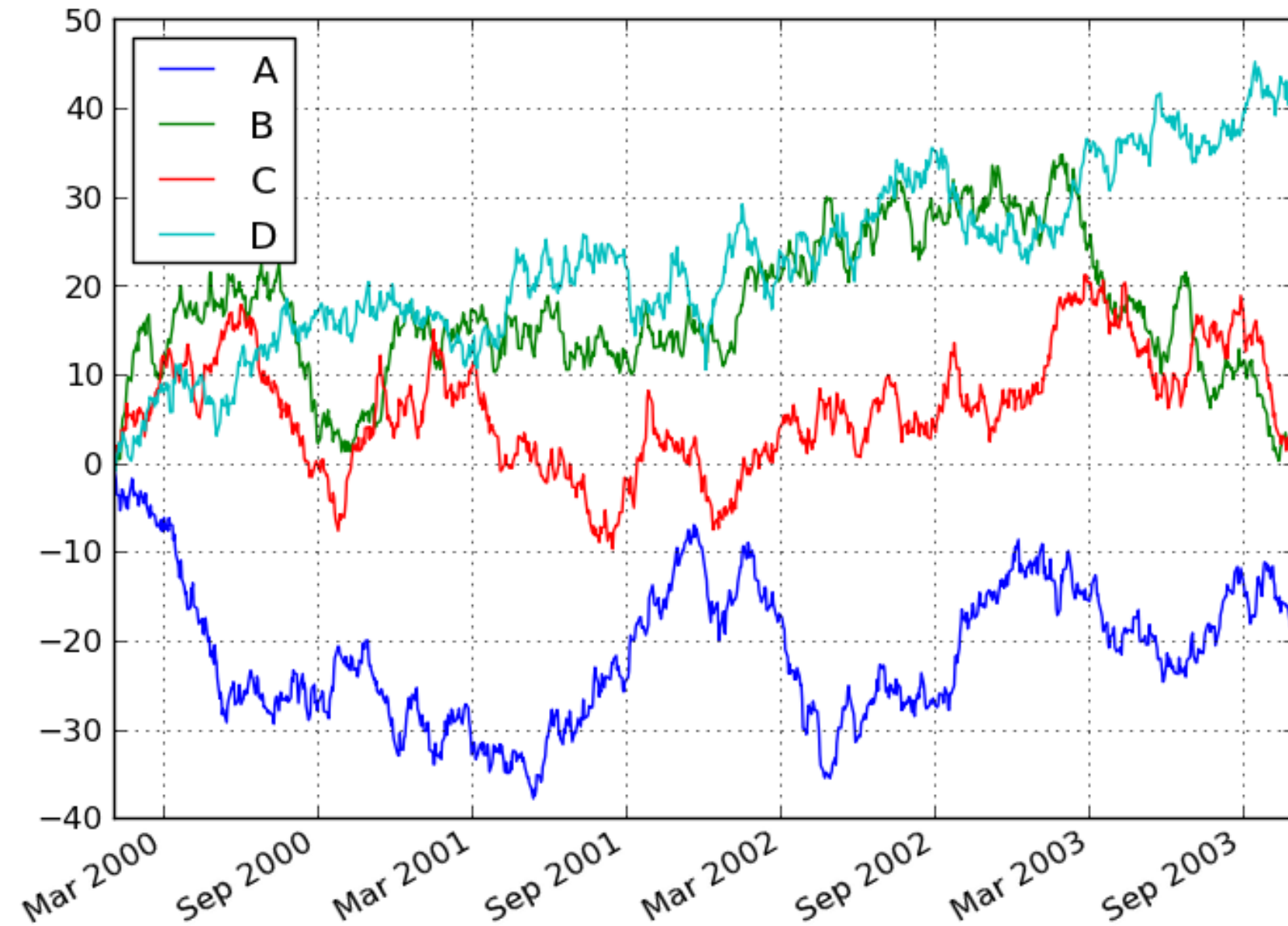
```
print(arr[0])
```

```
2
```

```
print(arr[-1])
```

```
32
```

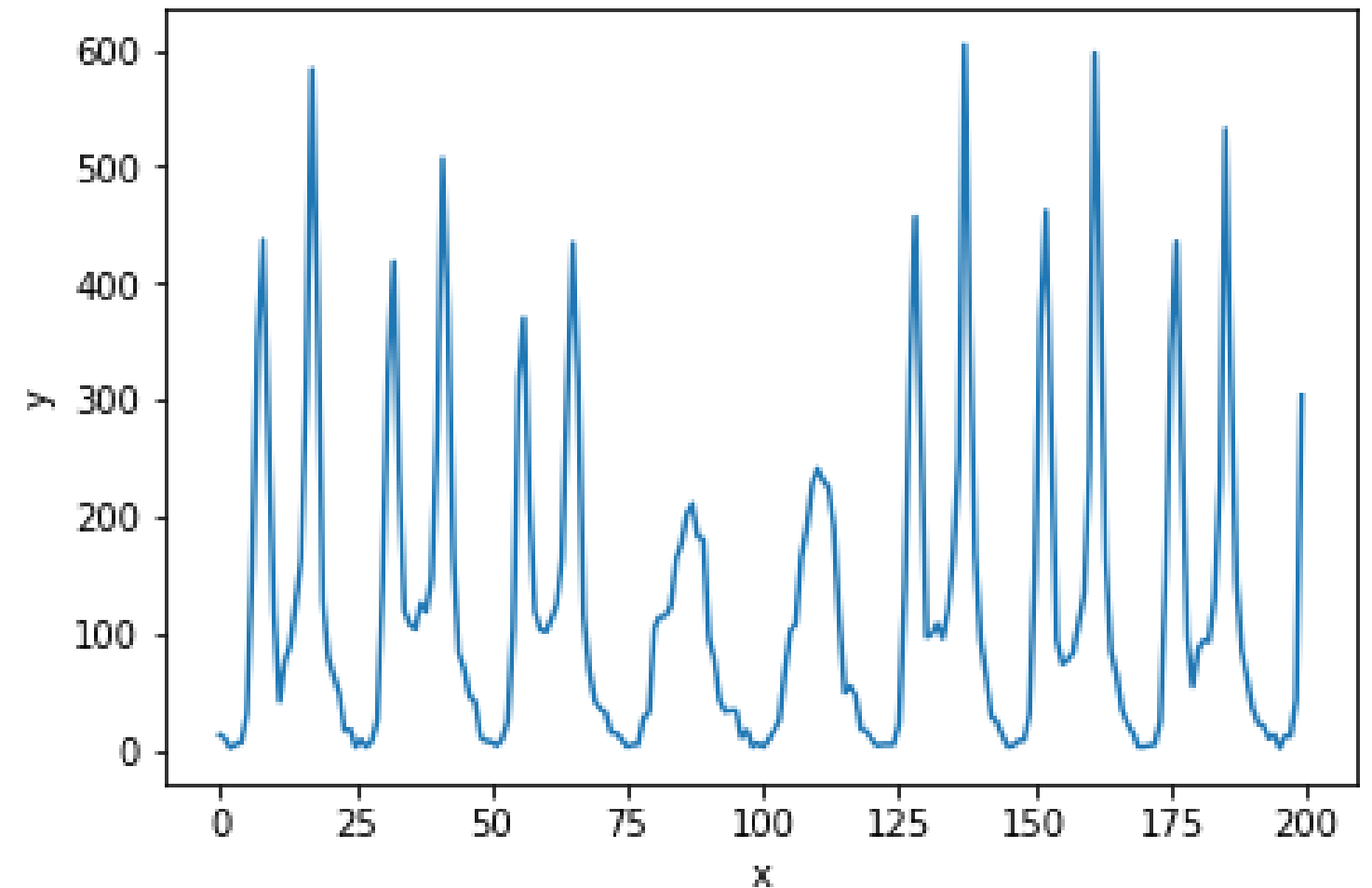
Visualizing data with Matplotlib



Using matplotlib

```
import matplotlib.pyplot as plt
```

```
# Create a new figure
plt.figure()
# Plot y as a function of x
plt.plot(x, y)
# Set the x-label and y-label
plt.xlabel('x')
plt.ylabel('y')
# Be sure to show the plot!
plt.show()
```



Let's practice!

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